

1. GENERAL INFORMATION OF THE COURSE

Course Name: Electrical Machines 5
Course Code: ECEG-4241
Course Credit Points: 5 ECTS
Contact Hours: 5 hrs/week
Course Module: Machines and Drives
Course Category: Core
Status of Course: Compulsory
Pre-Requisites: Introduction to Electrical Machines
Students' Mechanical Engineering
Degree Program: Bachelor of Science
Year of Study: 3
Semester: I
Academic Year: 2025
Course Duration: One semester
Module Coordinator: N.N.
Lecturer: Getahun D
Consultation: To be announced

2. COURSE DESCRIPTION

This course provides advanced knowledge of electrical machines including transformers, DC machines, synchronous machines, and single-phase motors. The course emphasizes construction, operating principles, modeling, control aspects, and dynamic behavior of these machines. Students will also gain exposure to fractional horsepower motors and modern motor applications in electrical power systems.

3. OBJECTIVES OF THE COURSE

On completion of this course the students are able to:

- ✓ Explain the different types of transformers with their connections.
- ✓ Develop D.C. armature winding & A.C. windings with their design aspects.
- ✓ Explain dynamic equations and control aspects of D.C. machines.
- ✓ Explain salient pole synchronous machine features, reference frame transformation, d-q axis theory,

power/load angle relationship, and carry out transient analysis.

✓ Explain the principle of operation and construction of fractional horsepower motors.

4. THE COURSE TIME SCHEDULE AND TEACHING METHODOLOGY

Weeks	Course Contents	Teaching & Learning Method
1–2	Introduction, basic working principles and construction of transformer Transformers: Parallel operation of single-phase transformers, Three-phase transformers and connections, Autotransformers.	Lectures, Tutorials
3–4	Multi-winding transformers, Instrument transformers, Voltage & Current transformers. Lab: Transformer testing & parameter measurement.	Lectures, Lab Exercises
5–6	Introduction, basic working principles and construction of DC Generator – Methods of excitation, General features, Types of D.C. armature windings.	Lectures, Tutorials
7–8	DC Machines: Dynamics, Control aspects, Armature reaction & commutation, Methods of improving commutation. Lab: DC motor load test, Speed control of DC motors.	Lectures, Lab Exercises

9	DC Machines: Parallel operation of DC machines, Design aspects.	Lectures, Assignments
10–11	Introduction, basic working principles and construction of Synchronous Machines: Introduction, Poly-phase machines, Reference frame transformation, Salient-pole machine analysis using d-q axis.	Lectures, Tutorials
12	Synchronous Machines: Power-load angle relationships, Transients, Parallel operation of alternators. Lab: Synchronous machine characteristics.	Lectures, Lab Exercises
13	Synchronous Machines: Transient analysis, Voltage & frequency regulation, Machine design considerations.	Lectures, Presentations
14	Single-phase Motors: Single-phase induction motor, AC series motor, Universal motor. Lab: Motor testing & performance.	Lectures, Lab Exercises
15	Single-phase Motors: Repulsion motors, Synchronous motors, Brushed and Brushless DC motor.	Lectures, Tutorials
16	Single-phase Motors: Stepper motors, Hysteresis motors.	Lectures, Lab Exercises

Lab: Performance testing
of stepper & DC motors.

5. ASSESSMENT AND EVALUATION

Assessment Type	Mark Allotted	Content Coverage	Time Schedule
Assignments	10%	Chapters 1–8	Throughout semester
Tests	20%	Chapters 1–8	Weeks 4 & 10
Seminar/Project Presentation	20%	Selected Topics	Week 14
Final Exam	50%	All Chapters	Week 16

6. COURSE POLICY

Students must adhere to the academic code of conduct. Academic dishonesty such as plagiarism and cheating will not be tolerated.

Attendance of at least 75% is mandatory to be eligible for the final exam. Students missing more than 25% of lectures will receive a grade of NG (No Grade) unless valid reasons are provided in line with university regulations.

7. GRADING SYSTEM

The grading scale follows the university's Senate Legislation Article 177 with fixed grading percentages for assignments, tests, seminar/project presentations, and final exams.

8. TEXTBOOKS

G. Mullisa: Introduction to Electrical Machines

9. REFERENCE BOOKS

- Kosow: Electric Machinery and Control, Prentice-Hall

- Siskind: Electrical Machines, McGraw-Hill
- Chapman: Electric Machinery Fundamentals, McGraw-Hill
- Fitzgerald, C. Kingsley, & S. D. Umans: Electric Machinery
- M.G. Say: Alternating Current machines
- J. Hindmarsh: Electrical Machines and their Applications

10. APPROVAL

Position	Name	Signature	Date
Course Instructor	Getahun D..	_____	2025
Course Chair	Atinafu	_____	2025
Chair Holder	Dr. Girmaw T.	_____	2025
V/Dean	Dr.Teketay M.	_____	2025